

Abylay Bissekenov

Astrophysicist

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Education

PhD **Xi'an Jiaotong-Liverpool University**, Physics — Suzhou, China
Mar 2024 – Dec 2026
- Thesis have been submitted on 27th of October
- Expected defence in November
- Diploma is expected by July of 2027

MS **Nazarbayev University**, Physics — Astana, Kazakhstan
Aug 2021 – June 2023
- GPA: 3.5/4.00
- Graduated with honors

BS **K.Zhubanov Aktobe Regional State University**, Physics — Aktobe, Kazakhstan
Sept 2016 – July 2020
- GPA: 3.64/4.00
- Graduated with honors

Experience

Xi'an Jiaotong-Liverpool University, Research Assistant — Suzhou, China
Mar 2024 – present
2 years 9 months

Xi'an Jiaotong-Liverpool University, Teaching Assistant — Suzhou, China
Mar 2024 – June 2026
2 years 4 months
- Over 450 hours of TAship experience

Energetic Cosmos Laboratory, Research Assistant — Astana, Kazakhstan
June 2022 – Dec 2024
2 years 7 months
- Worked under supervision of Bekdaulet Shukirgaliyev and Ernazar Abdikamalov on applications of ML techniques on N-body simulations.

Publications

Formation and Disruption of Wide Binaries in Star Clusters Revealed by N-body Simulations — Apr 2026
Abylay Bissekenov, Xiaoying Pang, Rainer Spurzem, Bekdaulet Shukirgaliyev, Mukhagali Kalambay, Peter Berczik
10.3847/2041-8213/ae5819 (The Astrophysical Journal Letters)

Evolution of star clusters with initial bulk rotation via N-body simulations — July 2025
Abylay Bissekenov, Xiaoying Pang, Kamlah Albrecht, M. B. N. Kouwenhoven, Rainer Spurzem, Bekdaulet Shukirgaliyev, Mirek Giersz, Abbas Askar, Peter Berczik
10.1051/0004-6361/202554093 (Astronomy and Astrophysics)

Cluster membership analysis with supervised learning and N-body simulations — Sept 2024
Abylay Bissekenov, Mukhagali Kalambay, Ernazar Abdikamalov, Xiaoying Pang, Peter Berczik, Bekdaulet Shukirgaliyev
10.1234/icml.2022.9012 (Astronomy and Astrophysics)

Conferences, Talks and Visits

- European Astronomical Society Annual Meeting 2026; Lausanne, Switzerland; 2026/06/29-2026/07/04 - K-Poster Presentation
- 1st Silk Road Conference; Tashkent, Uzbekistan; 2026/05/18-2026/05/22 - Contributed Presentation
- Structural, Formation and Evolution of Clusters 2026 Annual Symposium; Nanjing, China;2026/04/22-2026/04/26 - Contributed Presentation
- 2nd XJTU-XJTLU-UoL Doctoral Wisdom Convergence Camp; Xi'an, China; 2026/04/17-2026/04/19 - Oral Presentation
- 2026 XJTLU Postgraduate Research Symposium; Suzhou, China; 2026/03/19-2026/03/20 - Oral and Poster Presentation
- Informal Presentation at Fesenkov Astrophysical Institute; 2025/10/08 - Oral Presentation
- Informal Presentation at Al-Farabi Kazakh National University; 2025/10/07 - Oral Presentation
- Informal Presentation at Nazarbayev University; 2025/09/29 - Oral Presentation
- Galaxy Coffee Seminar at Max Plank Institute of Astronomy; 2025/07/29 - Oral Presentation
- Researcher visit to University of Heidelberg; Heidelberg, Germany; 2025/06/29-2025/08/04
- Galaxy Evolution Seminar of Eva K. Grebel group in University of Heidelberg; 2025/07/16 - Oral Presentation
- IAU Symposium 398/MODEST-25; Seoul, South Korea; 2025/06/15-2025/06/20 - Contributed Presentation
- 2025 XJTLU Postgraduate Research Symposium; Suzhou, China; 2025/03/19-2025/03/21 - Oral and Poster Presentation
- Structural, Formation and Evolution of Clusters 2024 Annual Symposium; Hangzhou, China; 2025/04/26-2025/04/29 - Contributed Presentation
- Academic mobility to Gdansk University; Gdansk, Poland; 2017/10/01-2018/01/25

Awards

- Best Oral Presentation Award - 2nd XJTU-XJTLU-UoL Doctoral Wisdom Convergence Camp (2026-04)
- Best Poster Presentation Award - XJTLU Postgraduate Research Symposium (2026-03)
- Excellent Oral and Poster Presentation Awards - XJTLU Postgraduate Research Symposium (2025-03)
- Deans List - Nazarbayev University (2023-09)

Skills

Languages: Kazakh, English, Russian

Programming Languages: Python, C/C++, HTML/CSS, Fortran

ML Frameworks: PyTorch, TensorFlow

Research Areas: Astrophysics, Physics, Data Science, High-Performance Computing